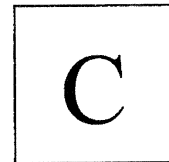


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B.Tech. Degree III Semester Supplementary Examination
April 2018

CS/IT 15-1304 OBJECT ORIENTED PROGRAMMING
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) How are data and functions organized in an object-oriented programme?
 (b) What do you mean by dynamic binding? How is it useful in OOP?
 (c) What is the main advantage of passing arguments by reference?
 (d) What is a class? How does it accomplish data hiding?
 (e) Why is it necessary to overload an operator?
 (f) What is the use of *this* pointer?
 (g) What is the difference between opening a file with constructor function and opening a file with open () function?
 (h) Describe various classes available for file operations.
 (i) Distinguish between class and function templates.
 (j) Why do we need virtual function?



PART B

(4 × 10 = 40)

- II. (a) Implement a C++ program to find the area of a rectangle using an inline function defined outside the class. (5)
 (b) Describe different parameter passing techniques with examples. (5)
- OR**
- III. (a) Explain the key concepts of OOP. (6)
 (b) Write a C++ program to exchange values between two classes using friend function. (4)
- IV. Define a class to represent a Bank account. Include the following members. (10)
 Data members: Depositor name, Account number, Type of account, Account Balance. Provide a default constructor, a parameterized constructor to this class. Also provide member functions:
 (i) To deposit amount.
 (ii) To display all the details of an account holder.
- OR**
- V. (a) Write a C++ program to count the number of persons inside a bank, by increasing count whenever a person enters a bank, using an increment (++) operator overloading function inside a class. (7)
 (b) Define type conversion. (3)

(P.T.O.)

- VI. (a) Explain any three types of inheritances with suitable examples. (5)
(b) Write a C++ program to find the minimum of 2 numbers using this pointer and return pointer reference. (5)

OR

- VII. (a) Create a base class *shape*, use this class to store two double type values that could be used to compute the area of figures. Derive two specific classes *triangle* and *rectangle* from the base *shape*. Add to the base class, a member function *getdata()* to initialize base class data members and another member function *displayarea()* to compute and display area of figures. Make *displayarea()* as a virtual function and redefine the function in the derived class to suit their requirements. (7)
(b) Explain what an abstract class is, and its applications. (3)
- VIII. (a) Write a program to search an element in a given set of elements using class template. (6)
(b) Write short notes on exception handling. (4)

OR

- IX. (a) Write a program to read the class object of student info such as name, age, sex, height and weight from the keyboard and to store them on a specified file using *read()* and *write()* functions. Again, the same file is opened for reading and displaying the contents of the file on the screen. (7)
(b) Write short notes on file modes. (3)
